Wenzhe Zhang, Ph.D. Candidate (Updated in Sep. 2026)

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Key words

Floating marine structures; Offshore renewable energy; Fluid-structure interactions; Shared mooring.

Work Experience

2024.09 – Now   **Teaching Assistant (80 h per year), Research Assistant**, Universidad Politécnica de Madrid.

Education

2023.09 – Now   **Ph.D. Universidad Politécnica de Madrid, Spain**
Naval architecture and ocean engineering
Supervisor: Associate Prof. Javier Calderon-Sanchez; Prof. Zhiyu Jiang

2026.08 – 2026.11   **Visiting Ph.D. Shanghai JiaoTong University, China**
Supervisor: Prof. Zhen Gao (Academician of the Norwegian Academy of Technological Sciences)

2025.05 – 2025.08   **Visiting Ph.D. University of Agder, Norway**
Supervisor: Prof. Zhiyu Jiang (Academician of the Norwegian Academy of Technological Sciences)

2020.09 – 2023.06   **M.Eng. Wuhan University of Technology, China**
Hydraulic engineering.
Supervisor: Prof. Jin Pan; Prof. Xiaobin Li

2016.09 – 2020.07   **B.Eng. Wuhan University of Technology, China**
Harbor, waterway, and coastal engineering.

Selected Research Publications

13 published SCI journal articles, **8** of them as First/Corresponding Author (*), **1** Highly Cited Paper
Google Scholar Citation: 321; h-index=8; i10-index=6 (2026.09)

Journal Articles

13. H. Peng, X. Li, **W. Z. Zhang***, J. Pan, and P. Hu, "Physics-informed multi-layer perceptron (MLP) model with data augmentation for ship-bridge collision safety assessment," **Ocean Engineering**, vol. 362, p. 126 339, 2026. DOI.

12. **W. Z. Zhang***, Z. Jiang, J. Calderon-Sanchez, *et al.*, "Wind-assisted propulsion systems for green shipping: A techno-economic review," **Ocean**, vol. 2, 2026. DOI.

11. R. Haider, W. Shi, Z. Lin, Q. Xiao, **W. Z. Zhang**, and X. Li, "Dynamic response of floating offshore wind turbine to focused wave excitation," **Physics of Fluids**, vol. 37, no. 8, 2025. DOI.

10. J. Pan, X. Y. Liu, **W. Z. Zhang**, and M. C. Xu, "Numerical study on the dynamical performance of multi-stable metamaterial structures under impact loading," **Ocean Engineering**, vol. 338, p. 121 938, 2025. DOI.

9. M. Song, J. Chen, **W. Z. Zhang***, J. Calderon-Sanchez, and Z. Y. Jiang, "Crashworthiness study of a monopile OWT with metamaterial-based fenders under vessel collisions," **Engineering Structures**, vol. 345, p. 121 559, 2025. DOI.

8. J. Tao, W. Chai, X. Yang, **W. Z. Zhang***, C. Wang, and J. Qi, "Stability evaluation of a damaged ship with ice accumulation in arctic regions," **Journal of Marine Science and Engineering**, vol. 13, no. 9, p. 1685, 2025. DOI.

7. **W. Z. Zhang**, J. Pan, H. Peng, and M. C. Xu, "Reliability assessment of aged reinforced concrete bridge under ship impact considering multi-hazards," **Ocean Engineering**, vol. 326, p. 120 772, 2025. DOI.

6. **W. Z. Zhang***, J. Calderon-Sanchez, T. Almeida-Medina, A. Medina-Manuel, and A. Souto-Iglesias, "Numerical analysis of forced heave oscillations for an FOWT column with a heave plate using overset mesh," **Ocean Engineering**, vol. 341, p. 122 736, 2025. DOI.

5. **W. Z. Zhang**, J. Pan, J. Calderon-Sanchez, X. B. Li, and M. C. Xu, "Review on the protective technologies of bridge against vessel collision," **Thin-Walled Structures**, vol. 201, p. 112 013, 2024. DOI.
4. **W. Z. Zhang***, J. Calderon-Sanchez, D. Duque, and A. Souto-Iglesias, "Computational Fluid Dynamics (CFD) applications in Floating Offshore Wind Turbine (FOWT) dynamics: A review," **Applied Ocean Research (Highly Cited Paper)**, vol. 150, p. 104 075, 2024. DOI.
3. J. Pan, **W. Z. Zhang**, Z. M. Sun, X. Qu, and M. C. Xu, "Experimental study on the dynamical response of elastic trimaran model under slamming load," **Journal of Marine Science and Technology**, pp. 1–16, 2023. DOI.
2. J. Pan, T. Wang, **W. Z. Zhang**, S. W. Huang, and M. C. Xu, "Study on the assessment of axial crushing force of bulbous bow for bridge against ship collision," **Ocean Engineering**, vol. 255, p. 111 411, 2022. DOI.
1. M. C. Xu, Z. J. Song, T. Wang, **W. Z. Zhang**, and J. Pan, "Empirical formulae assessment of ultimate strength for perforated stiffened panels under longitudinal compression," **Ocean Engineering**, vol. 264, p. 112 445, 2022. DOI.

Conference Proceedings

4. **W. Z. Zhang***, J. Calderon-Sanchez, and A. Souto-Iglesias, "Influence of diameters in double heave plates on hydrodynamic coefficients under constant material usage," in *International Conference on Offshore Mechanics and Arctic Engineering (OMAE2026)*, American Society of Mechanical Engineers, vol. 180583, 2026.
3. C. Wang, J. Pan, **W. Z. Zhang**, and M. Xu, "Study on dynamical response of improved cylindrical shell chain structure under impact load," in *Trends in Collision and Grounding of Ships and Offshore Structures*, CRC Press, 2025, p. 317.
2. J. Pan, H. Peng, J. Gan, **W. Z. Zhang**, and X. Li, "Study on crashworthiness performance of an airbag anti-collision device for vessel-bridge collision," in *International Conference on Offshore Mechanics and Arctic Engineering (OMAE2023)*, American Society of Mechanical Engineers, vol. 86847, 2023, V002T02A004.
1. **W. Z. Zhang**, J. Pan, N. Li, and M. Xu, "The safety assessment of ship-bridge collision based on a simplified dynamic model," in *International Conference on Offshore Mechanics and Arctic Engineering (OMAE2023)*, American Society of Mechanical Engineers, vol. 86847, 2023, V002T02A002.

Academic roles

2024-Now	**Journal peer reviewer** Over **50** reviews for top journals in the field of ocean engineering and offshore renewable energy. *Applied Energy, Marine Structures, Ocean Engineering, Engineering Conversion and Management, Applied Ocean Research, Journal of OMAE......*
2025-2027	**Master thesis reviewer & supervisor & committee & lecturer** *EMSHIP*, International Master program in Advanced Design of Sustainable Ships and Offshore Structures. *MAERM*, Master's Degree on Marine Renewable Energies Harnessing.

Selected Awards

2026	**National Award**, Chinese Government Award for Outstanding Self-Financed PhD Students Abroad.
	Journal Award, Excellent quality of Reviewing for the journal *Ocean Engineering*.
	45th OMAE Outreach Scholarship, Outreach program and financial aid by the 45th OMAE Conference. (Only 11 doctoral recipients from a global pool of 900+ applicants)
2025	**National Scholarship**, Financial Aid for Predoctoral Research Staff 2025 supported by Santander Scholarship.
2024	**National Thesis Award**, Outstanding Thesis in Hydraulic Engineering Majors of National Higher Education Institutions in China.
2023	**National Scholarship**, Spanish government FPI Scholarship for pre-doctor researcher.
	Excellent Thesis Award, Top 1% postgraduate thesis in Wuhan University of Technology.
	Excellent Postgraduate Award, Top 1% postgraduate in Wuhan University of Technology.

Selected Awards (continued)

2020 **Excellent Bachelor Award**, Top 1% bachelor in Wuhan University of Technology.

2019 **National 1st Level Award**, 8th China & International Marine Vehicle Design and Construction Contest.

2018 **National 2nd Level Award**, 7th China & International Marine Vehicle Design and Construction Contest.

Funding and projects

2025-2028 **Spanish National Plan Funding: RENEWFLOAT**, Budget: €125,000
Lead researcher, Design of shared mooring and multi-use floating renewable platforms, funded by the Ministry of Science, Innovation and Universities (MCIU) of Spain, PID2024-161775OB-C21.

2026-2027 **Shanghai Sci-tech Co-research Program: China-Spain collaboration**, Budget: ¥400,000
Lead researcher from Spanish side, Study on the Coupling Mechanism Between Fishing Gear Mesh Structure and Catch Escape Size Based on a Hydrodynamic Coupling Model and the Escape Threshold. Funded by Shanghai Municipal Government, 25HB2710100.

2023-2025 **Spanish National Plan Funding: UPM-FOWT-DAMP2**, Budget: €121,000
Lead researcher, Hydrodynamics of Motion Damping Devices for Floating Offshore Wind Turbines: Numerical modeling and small-scale experiments, funded by the Ministry of Science, Innovation and Universities (MCIU) of Spain, PID2021-123437OB-C21.

2024-2028 **European COST Project: Decarbonising Waterborne Transportation (DeWaTra)**
Participant, European Cooperation in Science and Technology (COST grant), CA23159, DeWaTra WG4: Renewable energy contributions to ship energy efficiency.

Skills

Languages **Mandarin Chinese**-Mother language; **English**-Fluent; **Spanish**-Basic

Computer skills **OpenFOAM** (CFD simulations of fluid-structure interaction, waves, FOWTs), **MATLAB** (Post-processing of data), **ANSYS LS-DYNA** (FEA simulations of impact and collision), **Solidworks & Rhino** (Construction of solid models), **LaTeX**, ...